

Frisch–Waugh–Lovell (FWL) Theorem: Detailed Proof

Set-up. Let $y \in \mathbb{R}^n$, $X_1 \in \mathbb{R}^{n \times k_1}$, $X_2 \in \mathbb{R}^{n \times k_2}$ with $[X_1 \ X_2]$ full column rank. The model is

$$y = X_1\beta_1 + X_2\beta_2 + u, \quad \mathbb{E}[u \mid X_1, X_2] = 0.$$

Define the projection and residual-makers on $\mathcal{C}(X_2)$:

$$P_{X_2} = X_2(X_2'X_2)^{-1}X_2', \quad M_{X_2} = I - P_{X_2}.$$

Recall $P_{X_2} = P_{X_2}' = P_{X_2}^2$, $M_{X_2} = M_{X_2}' = M_{X_2}^2$, and $P_{X_2}M_{X_2} = 0$.

Goal. Show that the OLS coefficient on X_1 from the full regression of y on $[X_1 \ X_2]$ equals the OLS coefficient from regressing $\tilde{y} = M_{X_2}y$ on $\tilde{X}_1 = M_{X_2}X_1$:

$$\hat{\beta}_1^{\text{full}} = \hat{\beta}_1^{\text{FWL}} = (X_1'M_{X_2}X_1)^{-1}X_1'M_{X_2}y.$$

Full OLS normal equations. OLS minimizes $\|y - X_1\beta_1 - X_2\beta_2\|^2$, yielding

$$X_1'(y - X_1\hat{\beta}_1 - X_2\hat{\beta}_2) = 0, \tag{1}$$

$$X_2'(y - X_1\hat{\beta}_1 - X_2\hat{\beta}_2) = 0. \tag{2}$$

From (2) we get

$$X_2'y = X_2'X_1\hat{\beta}_1 + X_2'X_2\hat{\beta}_2 \quad \Rightarrow \quad \hat{\beta}_2 = (X_2'X_2)^{-1}X_2'(y - X_1\hat{\beta}_1).$$

Plug into (1):

$$X_1' \left[y - X_1\hat{\beta}_1 - X_2(X_2'X_2)^{-1}X_2'(y - X_1\hat{\beta}_1) \right] = 0 \iff X_1'M_{X_2}(y - X_1\hat{\beta}_1) = 0,$$

hence

$$(X_1'M_{X_2}X_1)\hat{\beta}_1 = X_1'M_{X_2}y.$$

Because $[X_1 \ X_2]$ has full column rank, $X_1'M_{X_2}X_1$ is nonsingular, so

$$\boxed{\hat{\beta}_1 = (X_1'M_{X_2}X_1)^{-1}X_1'M_{X_2}y}.$$

Two-step residualization interpretation. Define residualized variables $\tilde{y} = M_{X_2}y$ and $\tilde{X}_1 = M_{X_2}X_1$. OLS of $\tilde{y}$ on $\tilde{X}_1$ solves $\tilde{X}_1'(\tilde{y} - \tilde{X}_1\beta_1) = 0$, i.e.

$$(X_1'M_{X_2}X_1)\beta_1 = X_1'M_{X_2}y.$$

Thus its solution coincides with the boxed expression above. Therefore

$$\hat{\beta}_1^{\text{FWL}} = \hat{\beta}_1^{\text{full}}.$$

Geometric view (optional). Decompose $y = P_{X_2}y + M_{X_2}y$. The component $P_{X_2}y$ lies in $\mathcal{C}(X_2)$ and is orthogonal to any vector in $\mathcal{C}(M_{X_2}X_1)$. Hence only $M_{X_2}y$ carries information about β_1 once X_2 is controlled for, which is why regressing $\tilde{y}$ on $\tilde{X}_1$ suffices.

Assumptions for identification and inference. Full column rank of $[X_1 \ X_2]$ ensures invertibility of $X_1' M_{X_2} X_1$; standard exogeneity and regularity give consistency and usual variance formulas, e.g.

$$\widehat{\text{Var}}(\hat{\beta}_1) = \hat{\sigma}^2 (X_1' M_{X_2} X_1)^{-1}, \quad \hat{\sigma}^2 = \frac{\|M_{[X_1 \ X_2]}y\|^2}{n - k_1 - k_2}.$$

□